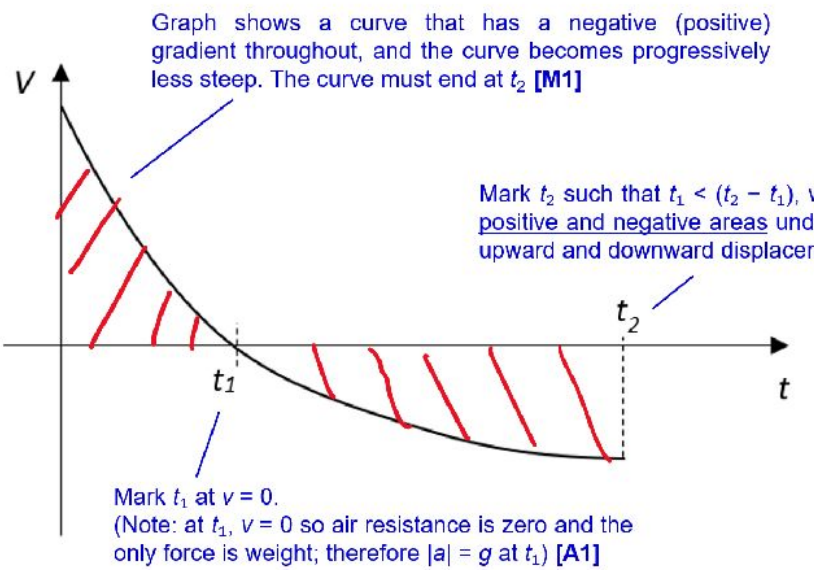
Qns	Answer	Marks
1(a)(1)	At t = 0, the gradient of the graph is 0. Hence the speed of the ball is 0.	A1
1(a)(2)	The gradient of the graph eventually becomes constant after the t = 0.80 s mark, indicating that the ball has reached terminal velocity. The acceleration, and thus the resultant force is zero. Since $F_{net} = mg - F_{air} = 0$, this shows that air resistance is significant and balances the weight when terminal velocity is reached.	M1 A1
1(b)	Draw a tangent to Fig. 1.1 at t = 0.40 s and determine its gradient. Speed = (2.8 ± 0.3) m s^{-1}.	M1 A1
1(c)		M1 A1 A1
	TOTAL FOR QUESTION 1	**8**
2(a)	The rate of change of momentum of a body is (directly) proportional to the resultant force	B1